

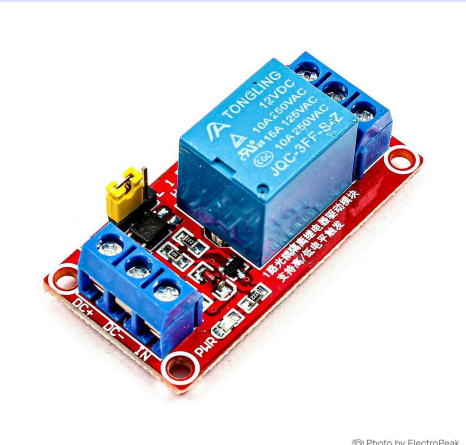
Smart Light Switch

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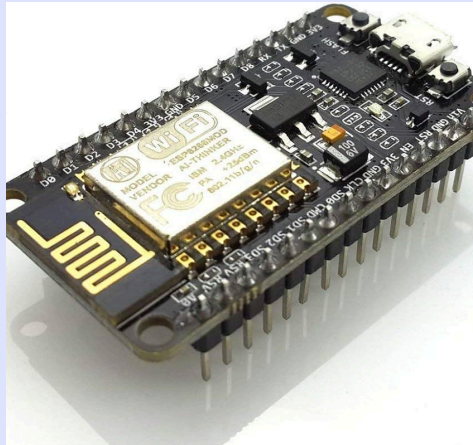
University: ESI

Components



Relay

Protecting the electrical system from too high of a voltage or current.



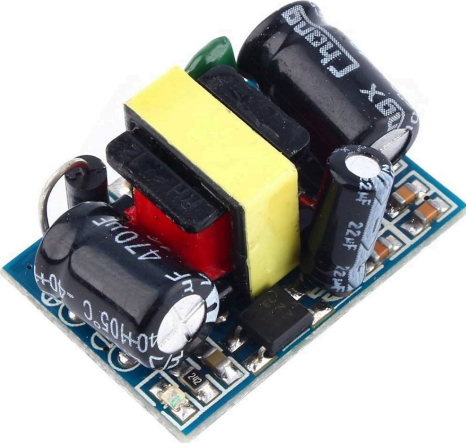
NodeMCU

Controlling various electronic devices and make them communicate with each other over a network.



Wall Outlet

allow electrical equipment to connect to the electrical grid.



Power Supply buck converter

converting a high voltage to a low voltage



Light bulb

produces light from electricity



PIR sensor grove

Sensing motion, usually human movement in its range



Description

The objective of this project is to develop an advanced smart switch that provides comprehensive control and automation capabilities for home lighting systems. The key features of this smart switch include remote control, voice control and motion-activated lighting.

Objectives

- Provide User-Friendly Interface.
- Optimize Cost-Effectiveness.
- Include Multi-User Support.
- Easy to use Product.

Process

Step 1: Gather all required components.

Step 2: Configure WiFiManager for Dynamic WiFi Connection.

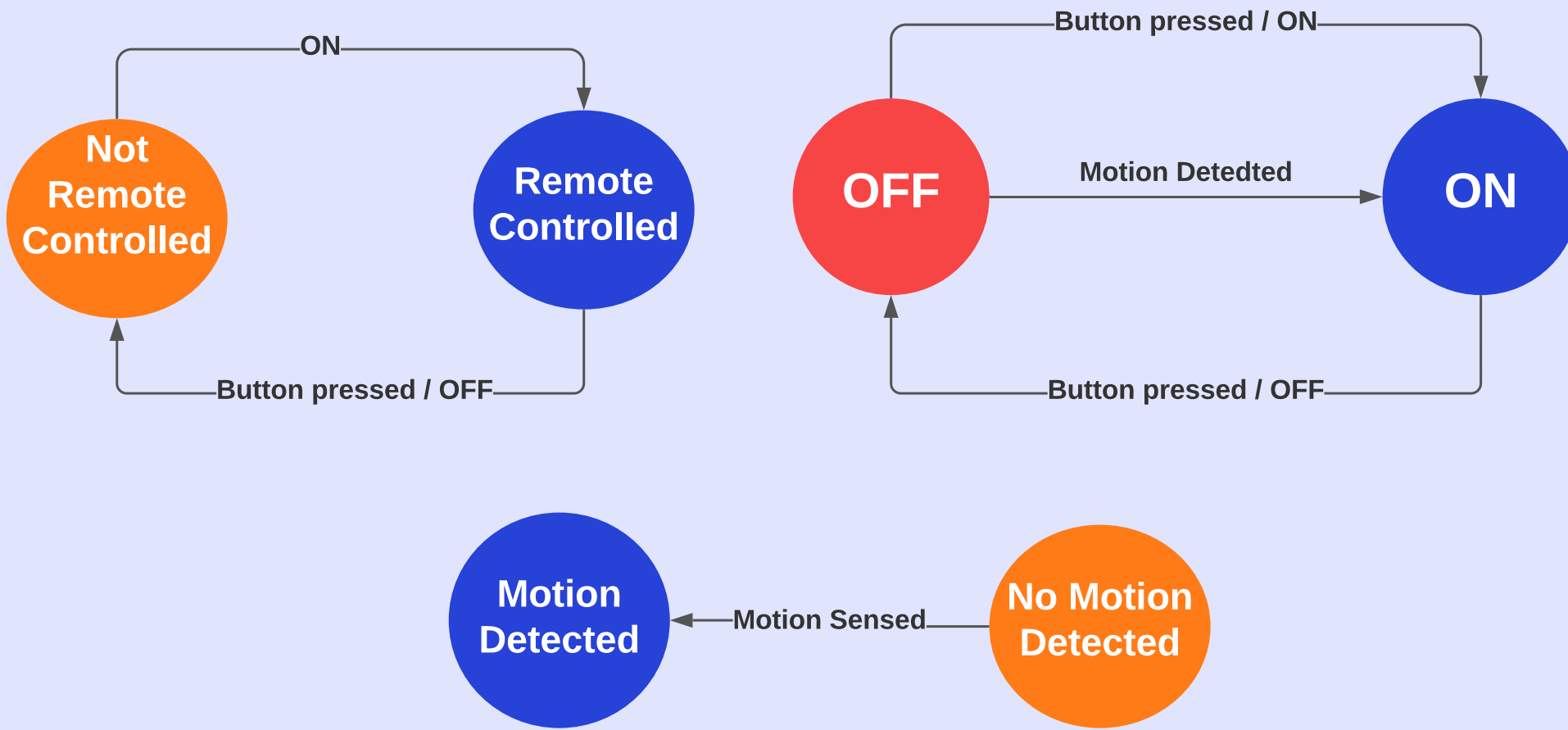
Step 3: Implement MQTT communication for light control.

Step 4: Motion Detection Logic.

Step 5: 3D Printing and Soldering.

Step 6: Connecting an Adapter (220V AC to 3.3V DC).

Step 7: Developing a react native app for voice and remote control.



State Diagram of the light and controll mode

Challenges

- WiFi management.
- Exlusive control.
- Synching.

Conclusion

Objective: Create an intelligent LED control system using NodeMCU with PIR motion sensor.

WiFiManager: Enabled flexible and user-friendly dynamic WiFi connectivity.

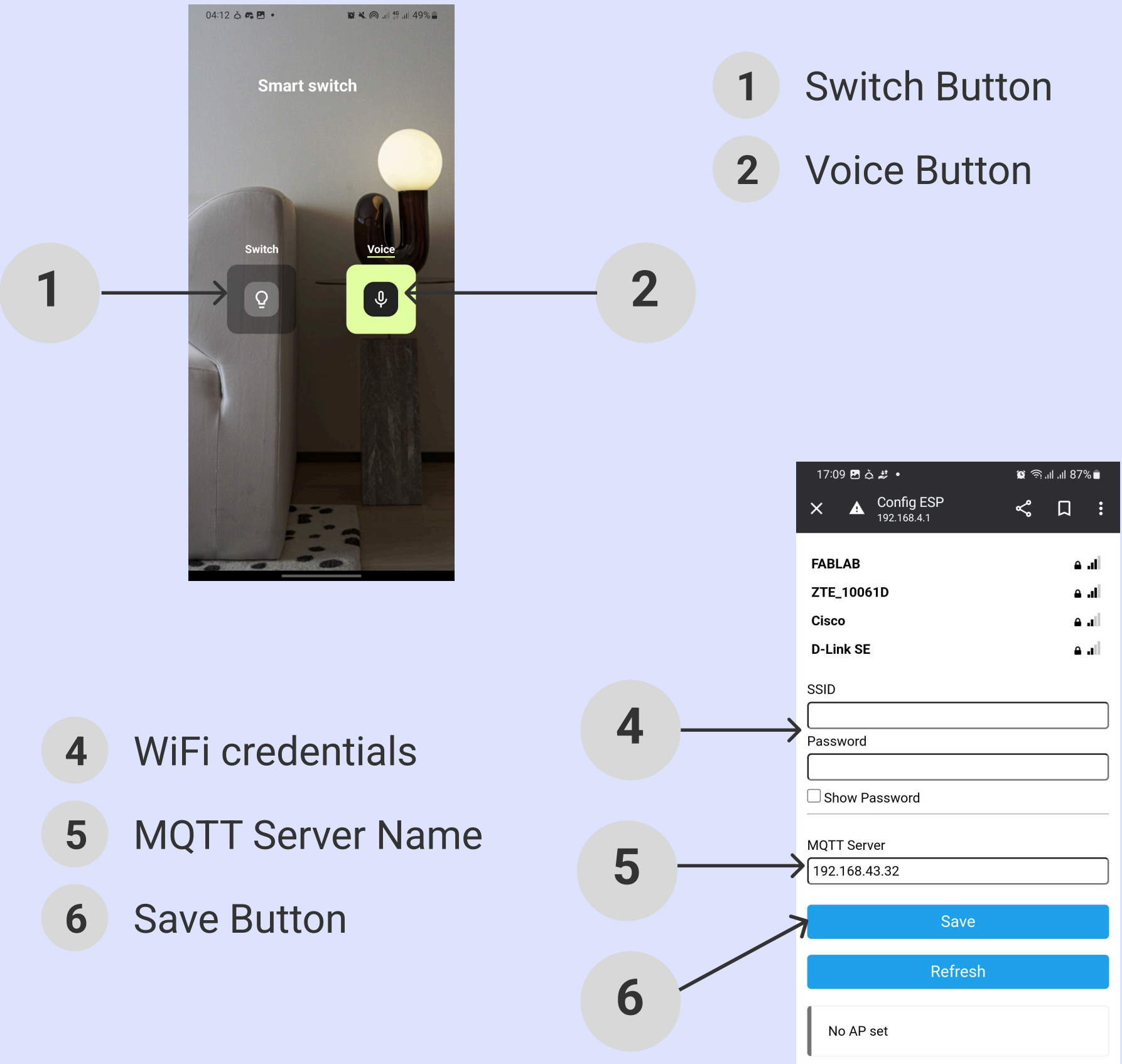
MQTT Communication: Provided reliable and efficient control of the LED based on motion detection and remote commands, ensuring exclusive functionalities.

3D Printing and Soldering: Added robustness and a professional finish to the setup.

Overall Achievement: Demonstrated how IoT technologies can create smart, interconnected devices that enhance convenience and automation.

Future Directions

- Multiple device control.
- Dimming.
- vacation Mode.



References

Horta, T. (2020, September 13). Mosquitto MQTT broker and ESP8266 communication in 6 easy steps. Medium. <https://tiagoorta1995.medium.com/mosquitto-mqtt-broker-and-esp8266-communcation-in-6-easy-steps-5ea0efae5a10>